# Band-1 Lonely Runner Finite-n Exact Rational Probe

Human + ZoraASI full-dash (proceed-dash)

2026-09-11

## Band-1 Computational Experiment: Lonely Runner Finite-$n$ Exact Rational Probe

**Honesty stamp.** This note does **not** claim that the Lonely Runner Conjecture (LRC) is solved, and does **not** claim a computer-assisted proof for unbounded speeds or for $n$ beyond tonight's bounded cells. It records a bounded exact-rational probe of integer speed tuples on this Linux box, plus a literature frontier ledger.

### Abstract

We execute Band-1 Lonely Runner as a finite verification probe. Using exact `fractions.Fraction` arithmetic, we evaluate loneliness at forbidden-arc boundary candidates and half-integer peaks on $\mathbb{R}/\mathbb{Z}$ for integer speeds. Results:

- Classical tight family $(1, 2, \dots, k)$ for $k = 1..12$ all attain loneliness $= \delta = 1/(k+1)$.
- **Ledger lean exhaustive** (this finalize run): zero failures for $(k, B) \in \{(1, 200), (2, 80), (3, 40), (4, 28), (5, 20),$ — **53,737** primitive tuples in $\approx 125.6$ s.
- **Session extended exhaustive** (earlier same-session long run; log-transcribed): zero failures for $(k, B) \in \{(1, 200), (2, 120), (3, 60), (4, 40), (5, 28), (6, 22), (7, 16)\}$ — **300,760** primitive tuples in $\approx 995.7$ s; deepening $(2, 200)$ added 12,231 more zero-failure tuples before abort.
- Eleven structured spot checks (powers-of-two, primes, Fibonacci, odd runs, classical extensions through $k = 12$) all pass.

Largest extended exhaustive cell this session: $k = 7$ **moving runners (**$n = 8$ **including stationary),** $B = 16$**, 11,432 primitive tuples, 0 failures**.

### Problem statement

Fix one runner stationary. Let $k$ be the number of moving runners with distinct nonzero speeds $v_1, \dots, v_k$. Loneliness of the stationary runner at time $t$ is

$$L(t) = \min_{1 \le i \le k} \|v_i t\|, \qquad \|x\| = \operatorname{dist}(x, \mathbb{Z}).$$

LRC asserts that for every choice of nonzero speeds there exists $t$ with $L(t) \ge \delta := 1/(k+1)$.

Tonight's probe restricts to **positive integer** speeds and asks only for bounded exhaustive certificates plus classical/spot witnesses — not a full proof for real speeds or unbounded integers.

## Method

For integer speeds the motion is 1-periodic. The closed good set

$$G = \{t \in \mathbb{R}/\mathbb{Z} : \min_i \|v_i t\| \geq \delta\}$$

has boundary among arc endpoints $t \equiv (j \pm \delta)/v_i \pmod 1$ and peaks at half-integers $v_i t \equiv 1/2 \pmod 1$. We evaluate $L(t)$ exactly (Fraction) at all such candidates and accept a tuple iff some candidate meets $L(t) \geq \delta$.

We check:

1. Classical family $(1, 2, \ldots, k)$ for $k = 1..12$.
2. All primitive strictly increasing positive integer $k$-tuples with max $\leq B_k$.
3. Structured spot checks beyond the exhaustive $B$ envelope.

Code: `/workspace/experiments/band1/lonely_runner_n/verify_lrc.py` and `run_ledger.py`.

## Results (numbers!)

**Classical family** $(1, 2, \ldots, k)$

| $k$ | $n$ | $\delta$ | witness $t$ | loneliness | pass |
|---|---|---|---|---|---|
| 1 | 2 | 1/2 | 1/2 | 1/2 | yes |
| 2 | 3 | 1/3 | 2/3 | 1/3 | yes |
| 3 | 4 | 1/4 | 1/4 | 1/4 | yes |
| 4 | 5 | 1/5 | 1/5 | 1/5 | yes |
| 5 | 6 | 1/6 | 1/6 | 1/6 | yes |
| 6 | 7 | 1/7 | 6/7 | 1/7 | yes |
| 7 | 8 | 1/8 | 1/8 | 1/8 | yes |
| 8 | 9 | 1/9 | 5/9 | 1/9 | yes |
| 9 | 10 | 1/10 | 3/10 | 1/10 | yes |
| 10 | 11 | 1/11 | 8/11 | 1/11 | yes |
| 11 | 12 | 1/12 | 11/12 | 1/12 | yes |
| 12 | 13 | 1/13 | 4/13 | 1/13 | yes |

All twelve classical rows attain equality loneliness $= \delta$.

**Ledger lean exhaustive (finalize run; zero failures)**

| $k$ | $n$ | $B$ | tuples checked | failures | seconds |
|---|---|---|---|---|---|
| 1 | 2 | 200 | 1 | 0 | 0.0001 |
| 2 | 3 | 80 | 1,965 | 0 | 4.5523 |
| 3 | 4 | 40 | 8,410 | 0 | 18.0867 |
| 4 | 5 | 28 | 19,343 | 0 | 43.3193 |
| 5 | 6 | 20 | 15,246 | 0 | 36.8835 |
| 6 | 7 | 16 | 7,980 | 0 | 20.7347 |
| 7 | 8 | 12 | 792 | 0 | 1.9908 |

**Session extended exhaustive (log-transcribed; zero failures)**

Source log: `/workspace/experiments/logs/2026-09-11_lonely_runner.log`. Completed zero-failure exhaustive sweeps before a mid-deepening abort; numbers transcribed from that log and treated as session evidence, not a Clay/LRC claim.

| $k$ | $n$ | $B$ | tuples checked | failures | seconds |
|---|---|---|---|---|---|
| 1 | 2 | 200 | 1 | 0 | 0.0001 |
| 2 | 3 | 120 | 4,385 | 0 | 13.7356 |
| 3 | 4 | 60 | 28,876 | 0 | 88.1834 |
| 4 | 5 | 40 | 85,771 | 0 | 274.6833 |
| 5 | 6 | 28 | 96,151 | 0 | 317.4891 |
| 6 | 7 | 22 | 74,144 | 0 | 263.8533 |
| 7 | 8 | 16 | 11,432 | 0 | 37.7517 |

**Deepening (partial; aborted mid-pass)**

| $k$ | $n$ | $B$ | tuples | failures | seconds |
|---|---|---|---|---|---|
| 1 | 2 | 500 | 1 | 0 | 0.0001 |
| 2 | 3 | 200 | 12,231 | 0 | 68.0200 |

Deepening beyond $k = 2, B = 200$ was truncated after the primary extended frontier $n = 2..8$ completed with zero failures. Further $B$ growth is optional and not claimed as a new published frontier.

**Structured spot checks**

| speeds | $k$ | $\delta$ | witness $t$ | loneliness | pass |
|---|---|---|---|---|---|
| $(1, 2, 3, 4, 5, 6, 7, 8)$ | 8 | 1/9 | 5/9 | 1/9 | yes |
| $(1, 2, 3, 4, 5, 6, 7, 8, 9)$ | 9 | 1/10 | 3/10 | 1/10 | yes |
| $(1, 2, 4, 8, 16)$ | 5 | 1/6 | 11/32 | 1/4 | yes |
| $(2, 3, 5, 7, 11)$ | 5 | 1/6 | 1/4 | 1/4 | yes |
| $(1, 3, 5, 7, 9, 11)$ | 6 | 1/7 | 1/2 | 1/2 | yes |
| $(1, 2, 3, 5, 8, 13)$ | 6 | 1/7 | 1/6 | 1/6 | yes |
| $(1, 2, 3, 4, 5, 6, 7)$ | 7 | 1/8 | 1/8 | 1/8 | yes |
| $(3, 5, 7, 9, 11, 13, 15)$ | 7 | 1/8 | 1/2 | 1/2 | yes |
| $(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)$ | 10 | 1/11 | 8/11 | 1/11 | yes |
| $(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11)$ | 11 | 1/12 | 11/12 | 1/12 | yes |
| $(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12)$ | 12 | 1/13 | 4/13 | 1/13 | yes |

## Literature context (ledger only — not re-proved here)

As of 2026-09-11:

- Classical analytic/combinatorial results: LRC for $n \leq 7$ runners (pre-2025 line).

- Rosenfeld (arXiv:2509.14111, 2025): computer-assisted proof for **eight runners**.
- Rosenfeld (arXiv:2512.01912, 2025): **nine runners**.
- Trakulthongchai (EJC 2026 / arXiv:2511.22427): CA proofs for **nine and ten runners**.
- Sungkawichai–Trakulthongchai (arXiv:2604.23906): CA proofs for **eleven–thirteen runners** ($k \leq 12$).
- Fourteen-runner CA extension (arXiv:2609.02604, 2026): **fourteen runners** ($k = 13$).

Standing published CA frontier: $n \leq 14$. Open beyond: $n \geq 15$.

**This Band-1 probe does not re-run those sieves** and does not certify unbounded-speed LRC($k$) from the $B_k$ cells alone. We do not stamp LRC closed.

## What this does not claim

- Not: LRC solved for all speeds / all $k$.
- Not: a finite-checking theorem that reduces unbounded integer speeds at these $k$ to the $B_k$ we scanned.
- Not: reproduction of Rosenfeld / Trakulthongchai / Sungkawichai modular sieves.
- Not: any Clay / Millennium stamp (LRC is not a Clay problem; listed only to keep the honesty gate consistent with the rest of the dash).

## Artifacts

| Path | Role |
|---|---|
| `/workspace/experiments/band1/lonely_runner_exact_verify_probe.py` | Exact verify probe |
| `/workspace/experiments/band1/lonely_runner_finalize_ledger.py` | Finalize ledger runner |
| `/workspace/experiments/band1/lonely_runner_finalize_ledger.json` | Machine-readable results |
| `/workspace/experiments/logs/2026-09-11_lonely_runner_extended_sweep.log` | Extended sweep console log |
| `/workspace/experiments/logs/2026-09-11_lonely_runner_finalize_ledger.log` | Finalize ledger log |
| This note | Human-readable certificate |

Finalize ledger wall: $\approx 125.6$ s. Extended exhaustive accounted wall: $\approx 995.7$ s (+ deepening $\approx 68$ s before abort).

## Honesty / dash gates

GATE-04 collection remains **LOCKED** (Object 1 apparatus still open). No invented $J$. No finished-ToE claim. No Clay-closed stamp. Band-1 Lonely Runner item: **bounded finite probe complete**; conjecture remains open for $n \geq 15$ and for unbounded speeds at every $k$.